Software Test Plan (STP)

Hospital Management System

Project	Hospital Management System
Version	1.0
Prepared by	QA / Project Team
Date	14-11-2026
Status	Updated working test plan – execution pending
Basis	Supplied repository, use case flow, SRS workbook and 100-case test-plan workbook

1. Introduction

1.1 Purpose

This Software Test Plan defines the test scope, objectives, strategy, environment, responsibilities, schedule, deliverables, risks and traceability approach for the Hospital Management System.

1.2 Scope

Testing covers patient registration and OTP verification, patient/doctor login, doctor listing and availability, appointment booking and cancellation, medical history, diagnosis, prescriptions, report upload/download, medicine ordering and demo payment, and bed availability/management.

1.3 References

- Hospital_Management_System_SRS.docx
- Hospital_Management_System_SAD.docx
- SRS_SE_PROJECT.xlsx
- Hospital_Management_System_Test_Plan.xlsx
- Hospital_Management_System_Use_Case_Flow.pdf
- Hospital_Management_System_System_Test_Report.pdf
- Hospital_Management_System_Unit_Test_Report.pdf

1.4 Definitions

UT – Unit Test; IT – Integration Test; ST – System Test; OTP – One-Time Password; JWT – JSON Web Token; RTM – Requirements Traceability Matrix.

2. Test Items

- React frontend application (React 19 / Vite 8)
- Node.js / Express 5 backend API
- Authentication and OTP workflows
- Appointment and bed management modules
- Medical history, diagnosis, prescriptions and report handling

- Medicine ordering, bills and demonstration payment workflows

3. Features to be Tested

- Patient registration, validation, duplicate rejection, OTP expiry/attempt handling and login
- Doctor login, role restrictions and assigned-appointment access
- Doctor directory, date/slot validation, booking, listing and cancellation
- Patient medical history, doctor diagnosis and prescription workflows
- Medical report metadata, file upload validation, ownership checks and download
- Prescribed medicine listing, order creation, demo payment, bills
- Bed listing, ward filtering, allocation and release
- API validation, error responses, authentication, authorization and application health

4. Features Not to be Tested / Out of Scope

- Actual hospital hardware and medical device integrations; none are present in the supplied source.
- Production SMS delivery; the current OTP is logged to the development console and no SMS provider is configured.
- Real financial settlement through an external payment gateway; the project uses a demo payment endpoint.
- Production database failover, backup restoration and transaction durability; the current backend uses in-memory arrays.
- Formal regulatory certification or clinical validation of medical advice and reports.

5. Test Approach / Strategy

5.1 Test Levels

Level	Purpose	Coverage
Unit / Component	Verify individual UI components and backend logic in isolation.	Form validation, rendering, slot rules, prescription validation, bed state transitions.
Integration	Verify communication and data flow between frontend, API middleware and backend modules.	Registration-to-OTP, appointment booking, doctor-to-patient record linkage, prescription retrieval, file download.
System	Verify end-to-end user workflows through the application.	Registration, booking/cancellation, diagnosis/prescription, purchase/payment, bed checks.
Acceptance	Confirm the system meets agreed project requirements.	Requirement coverage, high-priority workflow verification and stakeholder sign-off.

5.2 Test Types

- Functional and negative testing
- Authentication and authorization testing
- Input validation and boundary testing
- File upload/download security checks
- Usability and UI workflow checks
- Regression testing after fixes

- Basic API availability and response checks

5.3 Entry Criteria

- A stable frontend and backend build is available.
- Required environment variables are configured in backend/.env without committing secrets.
- Test accounts, doctor configuration and representative records are prepared.
- Frontend and backend can be started and the API health endpoint is reachable.
- Test data and expected outcomes are reviewed by the project team.

5.4 Exit Criteria

- All planned cases have an execution result or an explicit blocked/not-applicable status.
- All critical/high-priority defects are resolved or formally accepted with rationale.
- Failed cases have defect references and retest results.
- Requirement-to-test traceability is complete for in-scope requirements.
- Test summary report is reviewed and approved.

5.5 Security Validation

- Verify missing, invalid, expired and wrong-role JWT access is rejected.
- Verify patients cannot access other patients' appointments, records or reports.
- Verify doctor routes require doctor role and appointment assignment.
- Verify password hashing and OTP expiry/attempt limits.
- Verify upload MIME/type and size restrictions and private download authorization.
- Verify payment demo limitations are clearly documented; do not treat the demo endpoint as production payment validation.

6. Test Environment

Component	Environment / Tools
Frontend	React 19, Vite 8, browser; local development server.
Backend	Node.js, Express 5; local API on configured port (source defaults API client to localhost:5000).
Authentication	bcryptjs, jsonwebtoken; test JWT_SECRET and configured doctor credentials.
File handling	Multer; private-reports directory; test PDF/JPEG/PNG and invalid/oversized files.
Test data	Synthetic patient accounts, doctor accounts, appointment slots, diagnoses, prescriptions, report files, medicines, orders and bed records.
Tools	Browser developer tools, Postman or equivalent API client, terminal, Microsoft Excel/Word, and Vitest/React Testing Library where configured in the working test environment.

Important environment limitation: the supplied ZIP does not include a persistent database configuration. Core data is stored in memory and is reset when the backend process restarts. OTP and payment integrations are demonstrations, not live external services.

7. Test Schedule

Milestone	Planned Timing	Status
Review SRS/SAD and finalize test cases	To be scheduled	Planned
Prepare local test environment and data	To be scheduled	Planned

Execute unit/component test cases	To be scheduled	Pending
Execute API/integration test cases	To be scheduled	Pending
Execute end-to-end system tests	To be scheduled	Pending
Defect retesting and regression	After defect fixes	Pending
Acceptance review and sign-off	To be scheduled	Pending

8. Test Deliverables

- Software Test Plan (this document)
- 100 planned test cases (UT, IT and ST)
- Test data and environment configuration notes
- Execution logs and screenshots
- Defect reports and retest evidence
- Test summary report and final RTM

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	To be assigned	Maintain test plan, coordinate execution and prepare summary.
Test Engineer	To be assigned	Prepare test data, execute cases, capture evidence and report defects.
Developer	Project team	Support environment setup, investigate defects and deliver fixes.
Project Guide / Product Owner	To be assigned	Review scope, resolve priority questions and approve acceptance results.

10. Risks and Mitigation

Risk	Mitigation
In-memory data is lost on restart	Prepare repeatable seed/test data and document this limitation; use a persistent database in a future release.
OTP cannot be received through SMS	Use the development console OTP for local test only; mark external SMS delivery out of scope.
Demo payment is not a real gateway	Test only the implemented demo behaviour; do not claim real settlement.
Local report directory unavailable or missing files	Prepare known test attachments and verify missing-file error handling.
Environment variables are missing or invalid	Provide a .env.example and verify JWT_SECRET/doctor configuration before execution.
Test case expectations differ from implementation	Record observed behaviour, raise requirement/design discrepancy and obtain approval before changing expected results.

11. Assumptions and Dependencies

- Node.js/npm and compatible browser are available.
- The backend .env file is configured locally; secrets are not included in the submitted ZIP.
- Doctor login values are configured through environment variables and map to a configured doctor ID.
- Test users and data are synthetic and contain no real patient information.

- The local development OTP is obtained from backend console output.
- External SMS, production payment, persistent database and hospital hardware are not available in the current project environment.

12. Suspension and Resumption Criteria

Suspend execution if the frontend/backend cannot start, the API health endpoint is unavailable, test data is corrupted, or a blocking defect prevents more than one-third of planned cases from being executed. Resume after the blocking defect is fixed, the environment is stable, and test data is restored.

13. Test Case Management and Traceability

The test case identifiers from the supplied workbook are retained. UT cases verify component-level behaviour, IT cases verify communication and cross-module data flow, and ST cases verify complete user workflows. Requirement references are mapped in the SRS RTM. Test Result is intentionally left as Pending until execution evidence is recorded.

14. Test Metrics and Reporting

- Execution progress = executed cases / total planned cases × 100
- Pass rate = passed cases / executed cases × 100
- Fail rate = failed cases / executed cases × 100
- Requirement coverage = requirements with at least one executed test / total in-scope requirements × 100
- Defect count by severity, open/closed status and age

Current plan baseline: 100 planned test cases from the supplied workbook. Actual Result and Test Result cells in that workbook are blank; therefore this document does not claim that these 100 cases have passed.

15. Approvals

Role	Name	Signature / Date
QA Lead	_____	_____
Development Lead	_____	_____
Project Guide / Product Owner	_____	_____

Appendix A – Detailed Test Case Catalogue (100 Cases)

The catalogue below preserves the IDs, modules, scenarios, preconditions, steps, test data and expected results from the supplied Hospital_Management_System_Test_Plan.xlsx. Execution result is Pending until the case is run and evidence is recorded.

Test ID	Module	Scenario	Preconditions	Steps / Test Data	Expected Result	Result
UT-01	Patient Registration & Login	Validate registration with all required patient details.	System is accessible and registration page is available.	1. Open the system. 2. Select Register. 3. Enter all required details. 4. Submit registration. | Data: Name: Ananya Rao; Age: 21; Sex: Female; Phone: 9876543210; Email: ananya.rao@example.com	Entered details are accepted and the registration process proceeds to OTP verification.	Pending
UT-02	Patient Registration & Login	Reject registration when a required field is missing.	Registration page is available.	1. Select Register. 2. Leave Email blank. 3. Enter the remaining required details. 4. Submit. | Data: Email: blank; other fields valid	The system validates the input and prevents registration until the required field is provided.	Pending
UT-03	Patient Registration & Login	Authenticate a patient using valid login credentials.	A patient account has already been created and verified.	1. Open login. 2. Enter valid credentials. 3. Submit. | Data: Registered phone/email	The system authenticates the patient and provides access.	Pending

				and correct password		
UT-04	Patient Registration & Login	Reject login with invalid credentials.	A patient account exists.	1. Open login. 2. Enter an incorrect password. 3. Submit. | Data: Valid username/phone; Password: incorrect123	The system displays an error and asks the patient to enter credentials again.	Pending
IT-01	Patient Registration & Login	Verify registration-to-OTP integration.	Registration page and OTP service are available.	1. Enter valid registration details. 2. Submit registration. 3. Observe the next step. | Data: Valid patient details	After validation, the system sends an OTP to the registered phone number and moves to OTP entry.	Pending
IT-02	Patient Registration & Login	Verify successful OTP verification creates the patient account.	Valid registration data has been submitted and a correct OTP has been received.	1. Enter the received OTP. 2. Submit verification. | Data: Correct OTP for the registered phone number	The OTP is verified and the patient account is created.	Pending
IT-03	Patient Registration & Login	Verify created account can be used for login.	Registration and OTP verification completed successfully.	1. Open login. 2. Enter the newly created account credentials. 3. Submit. | Data: Newly registered patient credentials	The system authenticates the new account and provides access.	Pending
ST-01	Patient Registration & Login	Complete the end-to-end patient registration and login flow.	System is accessible; OTP delivery is available.	1. Register with valid details. 2. Receive and enter OTP. 3. Complete account creation. 4. Log in with valid credentials. | Data: Valid patient details and received OTP	Registration, OTP verification, account creation, and login complete successfully.	Pending
ST-02	Patient Registration & Login	Verify invalid login is handled without granting access.	A registered patient account exists.	1. Open login. 2. Enter invalid credentials. 3. Submit. 4. Observe access state. | Data: Incorrect password	An error is displayed and protected patient access is not granted.	Pending
ST-03	Patient Registration & Login	Verify patient can retry login after an invalid attempt.	A registered patient account exists.	1. Enter invalid credentials and submit. 2. Correct the credentials. 3. Submit again. | Data: First attempt: incorrect password; second attempt: correct credentials	The invalid attempt is rejected and the subsequent valid attempt grants access.	Pending
UT-05	OTP Verification	Generate an OTP for a registered phone number.	A valid registered phone number is available.	1. Provide the registered phone number. 2. Request verification. | Data: Phone: 9876543210	The system generates and sends an OTP to the registered phone number.	Pending
UT-06	OTP Verification	Accept the correct OTP.	An OTP has been generated and received.	1. Enter the received OTP. 2. Submit. | Data: Correct received OTP	The OTP is validated and the phone number is verified.	Pending
UT-07	OTP Verification	Reject an incorrect OTP.	An OTP has been generated.	1. Enter an incorrect OTP. 2. Submit. | Data: OTP: 000000	The system rejects the OTP and asks the patient to enter the correct OTP.	Pending
UT-08	OTP Verification	Reject an empty OTP submission.	OTP entry screen is displayed.	1. Leave the OTP field empty. 2. Submit verification. | Data: OTP: blank	The system does not verify the phone number and prompts for a valid OTP.	Pending
IT-04	OTP Verification	Verify phone number is passed correctly from registration to OTP generation.	Patient registration has been submitted with a valid phone number.	1. Register with a valid phone number. 2. Trigger OTP verification. | Data: Phone: 9876543210	The registered phone number is used for OTP generation and delivery.	Pending
IT-05	OTP Verification	Verify generated OTP is accepted by the verification component.	An OTP has been generated for the patient.	1. Retrieve the received OTP. 2. Enter it. 3. Submit. | Data: Received OTP	The verification component validates the generated OTP successfully.	Pending
IT-06	OTP Verification	Verify successful verification permits continuation to login/registration.	Correct OTP is available.	1. Enter correct OTP. 2. Submit. 3. Continue to the next step. | Data: Correct OTP	The phone number is marked verified and the patient is allowed to continue.	Pending
ST-04	OTP Verification	Complete the end-to-end OTP verification flow.	System and OTP delivery are available.	1. Provide registered phone number. 2. Request OTP. 3. Enter received OTP. 4. Submit. | Data: Valid registered phone number and correct OTP	OTP is sent, validated, and the phone number is verified.	Pending
ST-05	OTP Verification	Verify incorrect OTP does not allow continuation.	An OTP has been generated.	1. Enter an incorrect OTP. 2. Submit. 3. Observe the next-step access. | Data: Incorrect OTP: 123456	Verification is rejected and continuation is blocked until a correct OTP is entered.	Pending
ST-06	OTP Verification	Verify retry with the correct OTP succeeds after an incorrect attempt.	An OTP has been generated and received.	1. Enter incorrect OTP and submit. 2. Enter the correct OTP. 3. Submit again. | Data: First OTP: incorrect; second OTP: received correct OTP	The first attempt is rejected; the correct OTP then verifies the phone number.	Pending
UT-09	View Medical History	Display stored medical history for an authorized patient.	Patient account is logged in and has stored medical history.	1. Open medical history. 2. Observe the displayed information. | Data: Existing patient medical history	The system retrieves and displays the available medical information and reports.	Pending
UT-10	View Medical History	Handle a patient record with no stored medical history.	Authorized user is logged in; selected patient has no stored history.	1. Open medical history. | Data: Patient record with no history	The system displays the available information state without exposing unrelated records.	Pending
UT-11	View Medical History	Allow an authorized doctor to view patient history.	Doctor is logged in and authorized for the patient record.	1. Select the patient's medical history. 2. Observe the result. | Data: Authorized doctor account; existing patient record	The patient's stored medical history and reports are displayed.	Pending
UT-12	View Medical History	Deny access to medical history for an unauthorized user.	User is logged in but is not authorized to access the selected patient record.	1. Attempt to open the patient's medical history. | Data: Unauthorized user; protected patient record	Access is denied and the patient's medical information is not displayed.	Pending
IT-07	View Medical History	Verify login state is integrated with medical-history access.	A valid user can log in and has authorization.	1. Log in. 2. Open medical history. 3. Observe the retrieved record. | Data: Valid credentials; authorized patient	The authenticated user's authorization is recognized and the correct medical history is retrieved.	Pending
IT-08	View Medical History	Verify stored reports are retrieved with the patient's history.	Patient record contains medical history and reports.	1. Open the patient's medical history. 2. Review the returned information. | Data: Patient record with history and reports	Available medical history and reports are retrieved and shown together.	Pending
IT-09	View Medical History	Verify authorization check occurs before record retrieval.	User is logged in but unauthorized for the target patient.	1. Request the target medical history. 2. Observe whether protected data is returned. | Data: Unauthorized user; target patient record	Authorization fails and protected medical information is not returned.	Pending
ST-07	View Medical History	View medical history end-to-end as a patient.	Patient is registered, logged in, and has stored records.	1. Log in. 2. Select medical history. 3. Review displayed information. | Data: Valid patient account with stored history	The patient's available medical history and reports are displayed.	Pending
ST-08	View Medical History	View medical history end-to-end as an authorized doctor.	Doctor is logged in and authorized to access a patient.	1. Log in as doctor. 2. Select patient history. 3. Review displayed information. | Data: Authorized doctor account; patient record	The authorized doctor can view the patient's stored medical information.	Pending
ST-09	View Medical History	Verify unauthorized access is blocked end-to-end.	A protected patient record exists and an unauthorized account is available.	1. Log in as unauthorized user. 2. Attempt to access the patient's history. | Data: Unauthorized account; protected patient record	The system denies access and does not display the patient's medical information.	Pending
UT-13	Doctor Availability & Appointment Booking	Display doctors for a selected medical field.	Patient is logged in.	1. Select a medical field. 2. Observe the doctor list. | Data: Medical field: Cardiology	The system displays doctors available for the selected medical field.	Pending
UT-14	Doctor Availability & Appointment Booking	Display available appointment schedules for a selected doctor.	Patient is logged in and a doctor is available.	1. Select a doctor. 2. View appointment schedules. | Data: Doctor: Dr. A; field:	Available appointment schedules are displayed.	Pending

				Cardiology		
UT-15	Doctor Availability & Appointment Booking	Book an available appointment slot.	Patient is logged in and the selected slot is available.	1. Select a medical field. 2. Select a doctor. 3. Select an available slot. 4. Confirm booking. | Data: Doctor: Dr. A; Slot: 10:00 AM, 15-Sep-2026	The appointment is confirmed and stored; confirmation is displayed.	Pending
UT-16	Doctor Availability & Appointment Booking	Prevent booking when no appointment slot is available.	Patient is logged in and selected doctor has no available slots.	1. Select medical field. 2. Select doctor. 3. View schedules. | Data: Doctor with all slots occupied	The system informs the patient that no slot is available and displays other available schedules/doctors if present.	Pending
IT-10	Doctor Availability & Appointment Booking	Verify medical-field selection retrieves the correct doctors.	Patient is logged in; doctor data exists for multiple fields.	1. Select a medical field. 2. Observe returned doctors. | Data: Field: Cardiology	Only relevant available doctors are displayed for the selected field.	Pending
IT-11	Doctor Availability & Appointment Booking	Verify doctor selection retrieves current appointment schedules.	A doctor has configured appointment schedules.	1. Select the doctor. 2. View available schedules. | Data: Selected doctor with available and occupied slots	The system displays the doctor's current available appointment schedules.	Pending
IT-12	Doctor Availability & Appointment Booking	Verify booking stores the selected slot and returns confirmation.	Patient is logged in; selected slot is available.	1. Select doctor and slot. 2. Confirm booking. 3. Reopen booked appointments. | Data: Doctor: Dr. A; Slot: 10:00 AM	The selected appointment is stored and a confirmation is displayed.	Pending
ST-10	Doctor Availability & Appointment Booking	Book an appointment through the complete patient flow.	Patient account exists; doctor and appointment data are available.	1. Log in. 2. Select medical field. 3. Select doctor. 4. Select available slot. 5. Confirm booking. | Data: Cardiology; Dr. A; available slot	The appointment is successfully booked and confirmation is displayed.	Pending
ST-11	Doctor Availability & Appointment Booking	Verify unavailable-slot handling through the complete flow.	Patient is logged in; selected doctor has no free slots.	1. Select medical field. 2. Select doctor. 3. View schedules. | Data: Doctor with no available slots	The system informs the patient that no appointment slot is available and offers other schedules/doctors where available.	Pending
ST-12	Doctor Availability & Appointment Booking	Verify booked appointment persists after returning to the appointment section.	A patient has successfully booked an appointment.	1. Complete a booking. 2. Navigate away. 3. Reopen booked appointments. | Data: Previously booked doctor and slot	The booked appointment remains stored and is shown in the patient's appointments.	Pending
UT-17	Appointment Cancellation	Display a patient's booked appointments.	Patient is logged in and has at least one booked appointment.	1. Open booked appointments. 2. Observe the list. | Data: Patient with one or more appointments	The patient's booked appointments are displayed.	Pending
UT-18	Appointment Cancellation	Display details of the appointment selected for cancellation.	Patient has a booked appointment.	1. Open booked appointments. 2. Select an appointment to cancel. | Data: Appointment ID: APT001	The selected appointment details are displayed.	Pending
UT-19	Appointment Cancellation	Cancel a booked appointment after confirmation.	Patient has a booked appointment.	1. Select appointment. 2. Review details. 3. Confirm cancellation. | Data: Appointment ID: APT001; Confirm: Yes	The appointment status is updated to cancelled.	Pending
UT-20	Appointment Cancellation	Keep appointment active when cancellation is not confirmed.	Patient has a booked appointment.	1. Select appointment. 2. Review details. 3. Do not confirm cancellation. | Data: Appointment ID: APT001; Confirm: No	The appointment remains active.	Pending
IT-13	Appointment Cancellation	Verify cancellation updates the appointment record.	A booked appointment exists.	1. Select and confirm cancellation. 2. Reopen the appointment record. | Data: Appointment ID: APT001	The stored appointment status is changed to cancelled.	Pending
IT-14	Appointment Cancellation	Verify a cancelled appointment's slot becomes available again.	A booked appointment occupies a slot.	1. Cancel the appointment. 2. Check the doctor's available schedules. | Data: Appointment slot: 10:00 AM	The cancelled slot becomes available again.	Pending
IT-15	Appointment Cancellation	Verify the cancellation status is reflected in the patient's appointment list.	Patient has a booked appointment.	1. Cancel the appointment. 2. Return to booked appointments. | Data: Appointment ID: APT001	The appointment is shown with cancelled status.	Pending
ST-13	Appointment Cancellation	Cancel an appointment through the complete patient flow.	Patient is logged in and has a booked appointment.	1. Log in. 2. Open booked appointments. 3. Select appointment. 4. Confirm cancellation. | Data: Appointment ID: APT001	The appointment is cancelled successfully and its status is updated.	Pending
ST-14	Appointment Cancellation	Verify non-confirmed cancellation leaves the appointment unchanged.	Patient has an active appointment.	1. Open the appointment. 2. Select cancel. 3. Do not confirm. 4. Recheck appointment. | Data: Appointment ID: APT001; Confirm: No	The appointment remains active.	Pending
ST-15	Appointment Cancellation	Verify cancelled slot can be booked again.	An appointment occupies a slot and the patient can cancel it.	1. Cancel the appointment. 2. Open doctor availability. 3. Check the previously occupied slot. | Data: Cancelled appointment slot: 10:00 AM	The previously occupied slot is available for booking again.	Pending
UT-21	Medical Diagnosis & Reports	Display available patient medical history to the doctor.	Doctor is logged in and has access to a patient appointment/record.	1. Open the patient's appointment/record. 2. Review medical history. | Data: Patient record with existing history	The patient's available medical history is displayed.	Pending
UT-22	Medical Diagnosis & Reports	Enter and save an initial diagnosis.	Doctor is viewing an authorized patient record.	1. Enter initial diagnosis. 2. Save the record. | Data: Diagnosis: Suspected viral infection	The initial diagnosis is accepted and saved to the patient's record.	Pending
UT-23	Medical Diagnosis & Reports	Specify required reports or scans.	Doctor is viewing the patient's record.	1. Select reports/scans. 2. Specify the required items. 3. Save. | Data: Required report: CBC; Scan: X-ray	Required reports/scans are recorded with the patient record.	Pending
UT-24	Medical Diagnosis & Reports	Proceed with diagnosis when some patient information is unavailable.	Doctor is logged in; some patient information is unavailable.	1. Open the patient record. 2. Review available information. 3. Enter diagnosis. 4. Save. | Data: Available information only; Diagnosis: Initial assessment	The doctor can proceed using available information and enter the diagnosis.	Pending
IT-16	Medical Diagnosis & Reports	Verify patient history is available to diagnosis workflow.	Doctor is authorized and patient record contains history.	1. Open patient appointment. 2. Review retrieved history. 3. Enter diagnosis. | Data: Patient history; diagnosis text	Retrieved patient information is available before diagnosis entry.	Pending
IT-17	Medical Diagnosis & Reports	Verify diagnosis and required reports are stored in the same patient record.	Doctor has access to patient record.	1. Enter diagnosis. 2. Add required reports/scans. 3. Save. 4. Reopen record. | Data: Diagnosis: Viral infection; Report: CBC; Scan: X-ray	Diagnosis and required reports/scans are saved and associated with the correct patient record.	Pending
IT-18	Medical Diagnosis & Reports	Verify saved diagnosis can be retrieved after the consultation workflow.	A diagnosis has been saved for the patient.	1. Save diagnosis. 2. Close/reopen patient record. 3. Review diagnosis. | Data: Saved diagnosis: Initial assessment	The saved diagnosis is retrievable from the patient's record.	Pending
ST-16	Medical Diagnosis & Reports	Complete the doctor diagnosis and report workflow.	Doctor is logged in and an appointment/patient record exists.	1. Open patient appointment. 2. Review history. 3. Enter diagnosis. 4. Specify reports/scans. 5. Save. | Data: Diagnosis and required report/scan values	Diagnosis and required reports/scans are saved to the patient's record.	Pending
ST-17	Medical Diagnosis & Reports	Verify diagnosis can be completed with partial patient information.	Some patient information is unavailable.	1. Open patient record. 2. Review available information. 3. Enter diagnosis. 4. Save. | Data: Partial record; Diagnosis: Initial assessment	The doctor can proceed with available information and save the diagnosis.	Pending

ST-18	Medical Diagnosis & Reports	Verify stored diagnosis and reports are visible in the patient record after completion.	Doctor has completed and saved diagnosis/report details.	1. Save diagnosis and report requirements. 2. Reopen patient record. 3. Review stored information. | Data: Previously saved diagnosis and report/scan details	The saved diagnosis and required reports/scans remain associated with the patient record.	Pending
UT-25	Prescription Management	Allow doctor to enter final prescription and medication details.	Doctor is logged in and has opened the patient's medical record.	1. Enter final prescription. 2. Enter medication details. 3. Save. | Data: Prescription: Amoxicillin 500 mg; Frequency: 2 times/day	Prescription details are accepted for validation and storage.	Pending
UT-26	Prescription Management	Reject incomplete prescription details.	Doctor is logged in and prescription form is open.	1. Leave required medication information incomplete. 2. Attempt to save. | Data: Medication name: blank; dosage: 500 mg	The system asks the doctor to provide the required information before saving.	Pending
UT-27	Prescription Management	Display a stored prescription to the patient.	A valid prescription has already been stored for the patient.	1. Log in as patient. 2. Open prescription section. | Data: Stored prescription for patient	The stored prescription is displayed.	Pending
UT-28	Prescription Management	Validate and store complete prescription details.	Doctor is logged in and patient record is available.	1. Enter complete prescription details. 2. Save. 3. Reopen record. | Data: Medicine: Paracetamol 500 mg; Frequency: 3/day; Duration: 3 days	The prescription passes validation, is stored, and remains available in the record.	Pending
IT-19	Prescription Management	Verify prescription is associated with the correct patient record.	Doctor has access to a specific patient record.	1. Open patient record. 2. Enter prescription. 3. Save. 4. Reopen the record. | Data: Patient: P001; Prescription: Paracetamol 500 mg	The prescription is stored against the selected patient and not another record.	Pending
IT-20	Prescription Management	Verify stored prescription is retrieved by the patient-facing prescription section.	A prescription has been saved successfully.	1. Save prescription as doctor. 2. Log in as the patient. 3. Open prescriptions. | Data: Stored prescription	The patient-facing section retrieves and displays the saved prescription.	Pending
IT-21	Prescription Management	Verify incomplete prescription is not stored.	Doctor is logged in and form is open.	1. Enter incomplete details. 2. Attempt to save. 3. Reopen the patient record. | Data: Missing medication name	The incomplete prescription is rejected and is not saved as a valid prescription.	Pending
ST-19	Prescription Management	Complete prescription creation and patient viewing end-to-end.	Doctor and patient accounts exist; patient record is available.	1. Log in as doctor. 2. Open patient record. 3. Enter and save prescription. 4. Log in as patient. 5. Open prescription section. | Data: Complete prescription details	Doctor can save the prescription and the patient can view the stored prescription.	Pending
ST-20	Prescription Management	Verify incomplete prescription handling end-to-end.	Doctor is logged in and patient record is available.	1. Enter incomplete prescription. 2. Attempt to save. 3. Correct missing details. 4. Save again. | Data: First: missing medication name; second: complete details	The incomplete attempt is rejected; the corrected prescription is successfully stored.	Pending
ST-21	Prescription Management	Verify the stored prescription remains available after a new login session.	A prescription has been successfully stored.	1. Log out. 2. Log back in as patient. 3. Open prescription section. | Data: Previously stored prescription	The stored prescription is still displayed to the patient.	Pending
UT-29	Online Payment & Medicine Purchase	Calculate payment amount for selected prescribed medicines.	Patient has prescribed medicines available.	1. View prescribed medicines. 2. Select required medicines. 3. Observe calculated amount. | Data: Medicine A: quantity 1; Medicine B: quantity 2	The system calculates and displays the payment amount for the selected medicines.	Pending
UT-30	Online Payment & Medicine Purchase	Accept valid online payment details.	Patient has selected medicines and reached payment.	1. Select online payment. 2. Enter valid payment details. 3. Submit payment. | Data: Valid test payment details	The payment is processed securely and a successful payment confirmation is displayed when the payment succeeds.	Pending
UT-31	Online Payment & Medicine Purchase	Handle failed payment.	Patient has selected medicines and reached payment.	1. Enter payment details that cause a payment failure. 2. Submit. | Data: Payment data configured to fail in the test environment	The system displays payment failure and allows the patient to retry.	Pending
UT-32	Online Payment & Medicine Purchase	Record a successful medicine purchase.	A payment has completed successfully.	1. Complete successful payment. 2. Review purchase status/history. | Data: Successful medicine payment	The medicine purchase is recorded.	Pending
IT-22	Online Payment & Medicine Purchase	Verify selected medicines feed into payment calculation.	Prescribed medicines and prices are available.	1. Select one or more prescribed medicines. 2. Proceed to payment. 3. Compare selected items with calculated amount. | Data: Selected medicines with known test prices	The payment amount corresponds to the selected medicines.	Pending
IT-23	Online Payment & Medicine Purchase	Verify successful payment triggers purchase recording.	Medicine selection is complete and payment can succeed.	1. Submit valid payment. 2. Wait for success confirmation. 3. Check purchase record. | Data: Valid test payment details	Successful payment is confirmed and the medicine purchase is recorded.	Pending
IT-24	Online Payment & Medicine Purchase	Verify payment failure returns the user to a retry-capable state.	Medicine selection is complete.	1. Submit failing payment. 2. Observe failure response. 3. Attempt retry. | Data: Failing payment data, then valid test payment data	Failure is displayed and the patient is allowed to retry payment.	Pending
ST-22	Online Payment & Medicine Purchase	Complete medicine selection, payment, confirmation and purchase recording.	Patient has a valid prescription and prescribed medicines are available.	1. View medicines. 2. Select medicines. 3. Review calculated amount. 4. Select online payment. 5. Enter valid test payment details. 6. Submit. | Data: Prescribed medicines; valid test payment details	Payment succeeds, confirmation is displayed, and the medicine purchase is recorded.	Pending
ST-23	Online Payment & Medicine Purchase	Verify failed payment does not produce a false successful purchase.	Patient has selected medicines.	1. Submit a failing payment. 2. Observe result. 3. Check purchase record. | Data: Failing payment data	Payment failure is shown and a successful purchase is not recorded.	Pending
ST-24	Online Payment & Medicine Purchase	Retry a failed payment successfully.	Patient has selected medicines and the first payment attempt failed.	1. Submit failing payment. 2. Retry with valid test payment details. 3. Submit. | Data: First: failing test data; second: valid test payment data	The first attempt fails; the retry succeeds and the purchase is recorded.	Pending
UT-33	File Download	Display available medical documents.	Patient is logged in and documents exist.	1. Open medical documents section. 2. Observe available files. | Data: Reports and prescriptions stored for patient	Available reports, prescriptions and other files are displayed.	Pending
UT-34	File Download	Select an available file for download.	Medical documents section contains an available file.	1. Open documents. 2. Select a file. | Data: File: Prescription_P001.pdf	The selected file is identified and ready for download.	Pending
UT-35	File Download	Download an available medical file.	Patient is logged in and selected file is available.	1. Select the file. 2. Select Download. 3. Check the device download location. | Data: File: Report_P001.pdf	The selected file is downloaded to the patient's device.	Pending
UT-36	File Download	Handle a selected file that is unavailable.	Patient is logged in; requested file is unavailable.	1. Open documents. 2. Attempt to select/download the unavailable file. |	The system displays an appropriate error message.	Pending

				Data: Missing file reference		
IT-25	File Download	Verify medical-document records are connected to the patient's document list.	Patient has stored reports/prescriptions.	1. Log in. 2. Open medical documents. 3. Compare displayed files with stored documents. | Data: Stored report and prescription	Available stored documents are retrieved for the correct patient.	Pending
IT-26	File Download	Verify selected file is passed to the download function.	An available file is displayed.	1. Select a file. 2. Select Download. | Data: File: Prescription_P001.pdf	The requested file is passed to the download operation and download begins.	Pending
IT-27	File Download	Verify unavailable-file error is returned by the download operation.	A file reference exists but the underlying file is unavailable.	1. Select the unavailable file. 2. Select Download. | Data: Missing file reference	The system returns an appropriate error instead of downloading a nonexistent file.	Pending
ST-25	File Download	Download a medical report through the complete patient flow.	Patient is logged in and has an available report.	1. Open medical documents. 2. Select report. 3. Select Download. 4. Verify the downloaded file. | Data: Available report file	The selected report downloads successfully to the patient's device.	Pending
ST-26	File Download	Download a prescription through the complete patient flow.	Patient is logged in and has a stored prescription file.	1. Open medical documents. 2. Select prescription. 3. Select Download. 4. Verify the downloaded file. | Data: Available prescription file	The selected prescription downloads successfully.	Pending
ST-27	File Download	Verify unavailable-file handling through the complete flow.	Patient is logged in and selected file is unavailable.	1. Open medical documents. 2. Select unavailable file. 3. Select Download. | Data: Unavailable file	An appropriate error is displayed and no invalid file is downloaded.	Pending
UT-37	Bed Availability	Retrieve current bed information.	Bed availability section is accessible.	1. Open bed availability. 2. Observe retrieved bed information. | Data: Current bed dataset	The system retrieves current bed information.	Pending
UT-38	Bed Availability	Display available beds as available.	At least one required bed is available.	1. Open bed availability. 2. Observe bed status. | Data: Test data: required bed status = available	The system displays the bed as available.	Pending
UT-39	Bed Availability	Display occupied beds as occupied.	Required beds are occupied.	1. Open bed availability. 2. Observe bed status. | Data: Test data: required bed status = occupied	The system displays the bed as occupied.	Pending
UT-40	Bed Availability	Display all required beds as occupied when no beds are available.	No required beds are available.	1. Open bed availability. 2. Observe the availability message. | Data: All required beds = occupied	The system displays that all required beds are currently occupied.	Pending
IT-28	Bed Availability	Verify bed data retrieval feeds the availability check.	Current bed records exist.	1. Access bed availability. 2. Retrieve current bed information. 3. Check status. | Data: Current bed records	The availability check uses the current retrieved bed information.	Pending
IT-29	Bed Availability	Verify a bed-status update is reflected in the availability display.	A test bed record can be changed between available and occupied.	1. Set test bed status. 2. Refresh/open availability. 3. Observe status. | Data: Bed B001: available, then occupied	The displayed availability reflects the current stored status.	Pending
IT-30	Bed Availability	Verify the availability display distinguishes available and occupied states.	Test data contains both available and occupied beds.	1. Open bed availability. 2. Review multiple bed statuses. | Data: B001 available; B002 occupied	Each bed is displayed with the correct availability state.	Pending
ST-28	Bed Availability	Check bed availability end-to-end as a patient.	Patient can access the bed availability section; current bed data exists.	1. Log in. 2. Open bed availability. 3. Review status. | Data: Current bed data	The system displays current bed availability as available or occupied.	Pending
ST-29	Bed Availability	Check bed availability end-to-end as hospital staff.	Hospital staff account is available and current bed data exists.	1. Log in as staff. 2. Open bed availability. 3. Review status. | Data: Staff account; current bed data	The system displays current bed availability to authorized hospital staff.	Pending
ST-30	Bed Availability	Verify no-bed-available condition end-to-end.	No required beds are available in the test data.	1. Access bed availability. 2. Observe the displayed status/message. | Data: All required beds occupied	The system displays that all required beds are currently occupied.	Pending